

- 13** Catechol is a chemical found in a number of fruits. Catechol can be oxidised to a quinone by the enzyme catechol oxidase.

Catechol oxidase is inhibited by parahydroxybenzoic acid (PHBA) which is structurally similar to catechol.

Catechol oxidase is also inhibited by phenylthiourea (PTU) which binds to a copper atom in the enzyme.

How do both these inhibitors reduce the enzyme activity?

- 1 altering the specificity of the enzyme
- 2 competing with substrates for the active site
- 3 decreasing the $V_{\max}$ of the reaction

A 1, 2 and 3 **B** 1 only **C** 2 only **D** 3 only